Diana Dahyun Nam

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Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA

Education

- **Massachusetts Institute of Technology** *Sep 2026 - Present*
  Doctor of Philosophy in Mechanical Enginnering
- **Seoul National University** *Mar 2024 - Feb 2026*
  Master of Science in Mechanical Enginnering
- **Yonsei University** *Mar 2020 - Feb 2024*
  Bachelor of Science in Mechanical Enginnering

Research Experience

- **Design Computation and Digital Engineering (DeCoDE) Lab - MIT** *Sep 2026 - Present*
  Research Assistant (PI : Prof. Faez Ahmed)
  ◦ Developing robot co-design methods using agentic AI with hardware-in-the-loop implementation.
- **Biorobotics Laboratory - Seoul National University** *Sep 2023 - Mar 2026*
  UROP, Research Assistant (PI : Prof. Kyujin Cho)
  ◦ IMU-based robotic hip exoskeleton control system with human-in-the-loop optimization.
  ◦ Simulation and LQR-based controller for quasi-direct drive jumping-driving robot.
  ◦ Prototyped a fully foldable robotic wheel using composite manufacturing and custom electronics.
- **Soft Computing Lab - Yonsei University** *Mar 2023 - Aug 2023*
  Undergraduate Researcher (PI : Prof. Sung-Bae Cho, Prof. Seok-Jun Buu)
  ◦ Developed autonomous anomaly detection robot in manufacturing using multimodal DNN.
  ◦ In-vehicle noise classification by ProtoNet-based few-shot learning and dynamic memory replay.
- **Center for Healthcare Robotics - Korea Institude of Science and Technology(KIST)** *Dec 2022 - Feb 2023*
  Research Intern at Medical Navigation Lab (PI: Prof. Deukhee Lee)
  ◦ Developed vision-based complete blood count system using image processing and deep learning.
  ◦ Biological value estimation using optical microscope ray simulation and blood cell modeling.
- **Manufacturing and Mechatronics Lab - Yonsei University** *Sep 2022 - Dec 2022*
  Undergraduate Intern (PI: Prof. Byung-Kwon Min)
  ◦ Designed feedback control system of hardware-twin robot arm by two-way remote communication.

Work Experience

- **Hyundai Motors Group, Headquarters** *Jul 2022 - Oct 2022*
  Intern, S Innovation Progress Team, Robotics Division
  ◦ Robotics technology development, startup analysis, and patent assetization at HMGICS.

Publications

C=Conference, J=Journal, T=Thesis, P=In Preparation

[P.3] Dahyun Nam, Wonseok Choi et al. **Gram-Suit: A Sub-1 kg Soft Hip Exosuit with Extension-Dominant Bidirectional Assistance via a Single-Motor Differential Mechanism**. *(Science Robotics, Under Review)*

[P.2] Jinhee Lee, Dahyun Nam et al. **Vision-Ultrasound Robotic System based on Deep Learning for Gas and Arc Hazard Detection in Manufacturing**. *(Engineering Applications of Artificial Intelligence Under Review)*

[P.1] Seung-Hyun An, Sang-Min Baek, Dahyun Nam et al. (2026) **Thermally Limited Motor Overdrive for Continuous Quasi-Direct-Drive Jumping in a Small-Scale Wheeled Robot**. *(Journal of Field Robotics)*

[J.2] Dahyun Nam et al. (2024). **Design and Development of an Unmanned Water Rescue Platform with Emergency Detection and Self-inflating Life Jacket Launching Mechanism**. *Trans. Korean Soc. Mech. Eng. C*

[J.1] Robin Inho Kee, Dahyun Nam et al. (2024). **Disentangled Prototyping with Triplet-trained Prototypical Network for Few-shot Learning in In-vehicle Noise Classification**. *IEEE Access*

[C.2] Jegyeong Ryu, Dahyun Nam et al. (2023). **Corpuscular Volume Estimation Using a Small Amount of Blood via Microscope Images**. *2023 23rd International Conference on Control, Automation and Systems (ICCAS)*

[C.1] Dahyun Nam et al. (2023). **Dynamic Prototype-guided Memory Replay for In-Vehicle Noise Classification**. In *Conference of Korea Data Mining Society (KDMS)*

Honors and Awards

- **2026 Fulbright U.S.-Korea Presidential STEM Initiative Award** *Sep 2026 - Sep 2027*
  U.S. Fulbright Program
- **Outstanding MS Thesis Presentation Award** *Dec 2025*
  Department of Mechanical Engineering, Seoul National University
- **Excellent Teaching Assistant Award** *Mar 2025*
  College of Engineering, Seoul National University
- **BK21 FOUR Scholarship** *Mar 2024 & Mar 2025*
  Brain Korea 21 Program for Leading Universities and Students by the Korean government
- **The 13th KSME Student Creative Design Competition Excellent Project Award** *Sep 2023*
  The Korean Society of Mechanical Engineers (KSME)
- **SAS Student Paper Award** *Jun 2023*
  Conference of Korea Data Mining Society (KDMS)
- **Seoul Software Academy (SeSAC) Excellent Project Award** *Jun 2023*
  Seoul Business Agency
- **Creative Design Competition Grand Prize** *Jun 2023*
  Yonsei University College of Engineering

Extracurricular Activities

- **ETH Zurich 2025 Robotics Summer School in Epeisses, Switzerland.** *Jun 2025*
  Participated in the grant of ETH Robot X initiative fellowship
  ◦ Training on autonomous robot (state estimation, SLAM, planning, and object detection) and participated in search-and-rescue challenge.
- **Roboin, Robotics Club, Yonsei University** *Jun 2021 - Jan 2024*
  Executive Staff (2021-2024), Vice President (2022), President (2023-2024)
  ◦ Unmanned water rescue robot, quadruped robot, Teo-jansen linkage based robot and autorace project.
  ◦ Led robotics seminars(ROS, Embedded system, 3D printing) and cirriculums.
- **Yonsei Drone, UAV Club, Yonsei University** *May 2021 - Dec 2022*
  Executive Staff (2021-2022), Vice President (2021-2022), Advisory Committee (2022)
  ◦ Arduino based quadcopter design, position control and face-tracking drone project.
- **Undergraduate Internship, BAE, KIM & LEE LLC.** *Jul 2021*
  2021 Internship Program for Undergraduates, Labor & Employment team
- **Student Council of the College of Engineering, Yonsei University** *May 2021 - Dec 2021*
  Member of Communications Bureau and Freshman president of mechanical engineering

Teaching Experience

- **Teaching Assistant - Creative Design Engineering [400.018]** *2024 & 2025 Fall semester*
  Seoul National University, Dept. of Mechanical Engineering, Undergraduate Course [▶]
  ◦ Developed course materials on teleoperated mobile manipulator and delivered TA lectures.
- **Vex Robotics Instructor - Beyond Coding** *Jul 2024 - Nov 2024*
- **RC Makerspace Arduino Instructor - Yonsei Residential College** *May 2023*
- **Korean Literature Instructor - Insight Education Inc.** *May 2020 - May 2021*
- **Science Instructor - SUN Science Inc.** *Nov 2019 - May 2021*

Skills

- **Programming Languages:** C, C++, Python, Arduino, ROS
- **Design Tools:** Creo, Solidworks, AutoCAD, Fusion360, Onshape, Autodesk Inventor, KiCad
- **Mathematical & Simulation Tools:** MATLAB, Simulink, FEMM, Ansys Mechanical, TracePro, Isaac Sim, Mujoco

Certifications

- **Seoul National University, Dynamic Robot Systems Lab:** Robotics Boot Camp Certification *Jun 2022*
- **Yonsei University:** AI and Machine Learning Engineer Training Course Certification *Dec 2021*
- **Korea Transportation Safety Authority:** Qualification for Unmanned Multicopter Pilot 4th Class *Dec 2021*